

SKOPJE INTL, North Macedonia

WMO: 135860

Lat: 41.962N

Lon: 21.621E

Elev: 781

StdP: 14.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	17.6	5.6	7.5	14.6	11.4	10.1	21.4	21.6	41.2	18.7	36.7	2.1	50	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	97.0	68.4	93.8	68.1	91.2	67.4	71.5	89.6	70.2	87.6	69.0	85.9	5.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	99.4	76.1	64.6	94.0	74.8	63.4	90.1	74.1	35.6	89.8	34.5	88.1	33.6	86.0	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.1	16.4	13.7	DB	6.0	102.2	7.9	3.2	0.3	104.6	-4.3	106.4	-8.7	108.2	-14.4	110.6
			WB	5.3	74.4	7.6	1.5	-0.2	75.4	-4.7	76.3	-8.9	77.2	-14.5	78.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	33.5	38.6	46.4	54.5	63.3	71.1	75.5	75.6	66.6	56.4	45.4	35.6	(6)
	DBStd	16.13	8.09	7.80	6.96	6.08	5.49	5.70	4.88	4.82	6.02	6.84	7.53	8.51	(7)
	HDD50	1664	513	322	150	27	0	0	0	0	0	24	177	451	(8)
	HDD65	4551	977	738	577	315	98	14	1	2	53	275	589	912	(9)
	CDD50	3594	0	4	38	163	412	634	791	792	497	221	38	4	(10)
	CDD65	1007	0	0	0	1	45	198	327	329	100	7	0	0	(11)
	CDH74	12233	0	0	7	88	678	2348	3920	3911	1145	135	1	0	(12)
	CDH80	5680	0	0	0	15	173	1034	1990	2015	428	25	0	0	(13)
(14) Wind	WSAvg	3.7	3.0	4.0	4.7	4.3	3.7	4.0	4.7	4.0	3.6	2.9	2.8	3.0	(14)
(15) Precipitation	PrecAvg	20.4	1.3	1.1	1.4	1.9	2.0	1.9	1.4	1.2	1.8	2.2	1.8	2.4	(15)
	PrecMax	29.4	3.2	3.0	2.9	5.7	4.0	3.7	5.0	4.1	5.8	5.4	4.0	12.2	(16)
	PrecMin	12.2	0.1	0.2	0.1	0.6	0.9	0.5	0.0	0.0	0.2	0.0	0.0	0.1	(17)
	PrecStd	4.6	0.8	0.8	0.8	1.3	0.8	0.9	1.1	0.9	1.5	1.5	1.1	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	65.5	74.0	82.3	88.3	97.1	101.1	101.0	94.3	82.8	70.0	60.8	(19)
		MCWB	48.2	52.6	54.9	60.9	65.1	70.1	68.4	68.4	65.9	63.3	58.0	53.2	(20)
	2%	DB	51.6	59.4	69.3	75.9	84.4	93.1	97.0	97.1	89.4	77.4	64.4	53.8	(21)
		MCWB	46.1	49.2	53.3	58.5	64.5	69.0	68.6	68.2	65.5	60.8	56.4	49.8	(22)
	5%	DB	47.9	55.2	64.7	71.9	81.0	89.7	93.6	94.1	84.9	73.4	60.8	50.1	(23)
		MCWB	43.8	47.3	51.0	56.2	63.4	68.0	68.4	67.6	64.0	59.9	54.6	47.5	(24)
	10%	DB	44.4	51.2	60.7	68.4	77.4	86.2	90.6	91.0	80.7	69.5	57.3	46.7	(25)
		MCWB	41.6	45.2	49.4	55.0	62.1	67.0	67.6	67.0	63.0	58.9	52.3	44.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	53.7	56.7	62.8	68.6	73.0	72.9	73.3	69.2	65.4	59.9	54.5	(27)
		MCDB	54.2	63.2	70.1	77.1	82.3	90.2	90.5	91.5	84.4	77.2	65.6	59.8	(28)
	2%	WB	46.7	50.3	54.3	60.3	66.4	71.0	71.2	71.1	67.3	63.1	57.4	50.4	(29)
		MCDB	50.3	57.6	65.9	72.9	80.0	88.0	89.6	89.4	82.0	73.3	63.0	53.0	(30)
	5%	WB	43.9	48.1	52.2	58.1	64.7	69.5	70.0	69.9	65.9	61.3	55.3	48.1	(31)
		MCDB	46.8	54.3	62.5	69.4	78.0	85.7	88.2	87.0	79.9	70.2	60.3	50.3	(32)
	10%	WB	41.5	45.4	50.1	56.1	63.1	67.9	68.8	68.4	64.4	59.6	52.9	44.4	(33)
		MCDB	43.9	50.1	59.6	66.1	75.3	83.4	86.2	85.4	76.9	67.5	56.8	46.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	18.6	22.0	23.9	24.5	25.9	27.1	27.9	24.7	22.5	17.6	13.1	(35)
		MCDBR	19.4	26.9	31.9	31.5	31.2	31.4	33.0	33.9	32.4	29.6	22.4	16.6	(36)
		MCWBR	14.0	17.3	17.7	16.2	14.3	11.8	10.9	11.1	12.3	15.0	14.3	12.2	(37)
	5% WB	MCDBR	17.1	24.4	28.3	27.6	28.1	28.8	28.8	28.1	26.5	24.4	19.2	14.1	(38)
		MCWBR	12.8	16.2	16.3	14.9	13.9	11.7	10.4	10.1	11.0	13.3	12.5	10.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.334	0.360	0.405	0.463	0.456	0.435	0.437	0.454	0.432	0.395	0.362	0.335	(40)
	taud		2.353	2.271	2.185	2.038	2.128	2.235	2.226	2.174	2.213	2.300	2.347	2.356	(41)
	Ebn at Noon		253	264	264	256	262	267	265	256	252	248	239	241	(42)
	Edh at Noon		27	34	42	52	49	44	44	45	40	33	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		511	793	1121	1453	1768	2033	2120	1899	1372	960	610	437	(44)
(45)	RadStd		80	100	137	172	126	159	136	134	113	115	65	76	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-270	N/A	N/A

Nomenclature: See separate page